

Digital Image Representation

A **digital image** is a picture that is stored in a computer as **numbers**.

It is represented as a **matrix of pixels (picture elements)**.

Each **pixel** stores a value that represents the **intensity (brightness) or color** of that point in the image.

1. Mathematical Representation

A digital image can be represented as a **2D function**:

$$f(x, y)$$

Where:

- **x** → horizontal coordinate
- **y** → vertical coordinate
- **f(x, y)** → intensity value (brightness) at that pixel

If **x and y are discrete values**, the image becomes **digital**.

Example matrix representation:

120 125 130 140

115 120 128 135

110 118 125 130

Each number = **pixel intensity value**.

2. Pixel (Picture Element)

A **pixel** is the **smallest unit of a digital image**.

Example:

- Image size: **1024 × 1024**
- Total pixels: **1,048,576**

Each pixel stores **brightness or color information**.

3. Types of Digital Image Representation

(a) Binary Image

Pixel values:

- **0 → Black**
- **1 → White**

Example:

0 1 1 0
1 0 1 0
0 1 0 1

Used in **document scanning, OCR.**

(b) Grayscale Image

Pixel values range:

0 – 255

- **0 → Black**
- **255 → White**
- Middle values → shades of gray

Example:

0 50 120 255
30 90 160 200

(c) Color Image

Color images use **three channels**:

RGB (Red, Green, Blue)

Each channel value:

0 – 255

Example pixel:

R = 255

G = 0

B = 0

→ Red color

Total possible colors:

$$256^3 = 16,777,216$$

4. Resolution

Resolution means **number of pixels in the image**.

Example:

- **1920 × 1080**
- **1024 × 768**

Higher resolution → **more detail**.

5. Bit Depth

Bit depth defines **how many intensity levels** a pixel can store.

Bit Depth Intensity Levels

1 bit 2 levels

8 bit 256 levels

24 bit 16 million colors

Sampling and Quantization in Digital Image Processing

A **real-world image is continuous** in both **space and intensity**.

To store and process it in a computer, it must be converted into a **digital image**.

This conversion is done using two processes:

1. **Sampling** – digitizing the spatial coordinates
 2. **Quantization** – digitizing the intensity values
-

1. Sampling

Definition

Sampling is the process of converting the **continuous spatial coordinates (x, y)** of an image into **discrete coordinates**.

In simple words, it means **dividing the image into a grid of pixels**.

Mathematical Representation

A continuous image is represented as:

$$f(x, y)$$

Where

- **x, y** → spatial coordinates
- **f(x,y)** → intensity at that point

After sampling:

$$f(x, y) \rightarrow f(m, n)$$

Where

- **m, n** are **integer pixel locations**
-

Example

Suppose we have a **continuous image surface**.

After sampling it becomes a **matrix of pixels**:

120 135 140 150

110 125 130 140

100 115 120 135

Each number represents **pixel intensity**.

Sampling Rate

The **sampling rate** determines how many pixels are taken from the image.

Example:

Sampling Resolution Total Pixels

256 × 256	65,536
512 × 512	262,144
1024 × 1024	1,048,576

Higher sampling → **better image detail**

Effect of Low Sampling

If sampling is **too low**:

- image becomes **blurry**
- **loss of detail**
- **aliasing** occurs

Example: pixelated images.

Spatial Resolution

Sampling determines the **spatial resolution** of the image.

Spatial resolution means **how closely pixels are packed in the image**.

Higher spatial resolution → **more clear image**

2. Quantization

Definition

Quantization is the process of converting **continuous intensity values** into **finite discrete levels**.

In simple words, it assigns **brightness values to each pixel**.

Intensity Levels

Intensity values are usually stored using **bits**.

Example:

Bit Depth Number of Intensity Levels

1 bit	2 levels
2 bit	4 levels
8 bit	256 levels
16 bit	65,536 levels

Example

Suppose intensity values range from **0 to 255**.

Pixel Intensity Value

P1	25
P2	120
P3	200

These values represent **brightness of pixels**.

- **0 → Black**
- **255 → White**

Values between them represent **gray shades**.

Quantization Formula

If **b bits** are used, then number of intensity levels is:

$$L = 2^b$$

Where:

- **L** = number of intensity levels
- **b** = number of bits

Example:

8-bit image

$$L = 2^8 = 256$$

Quantization Error

During quantization, some information is lost.

This loss is called **quantization error**.

Example:

Actual intensity = **126.7**

Stored value = **127**

Difference = **0.3**

Effect of Low Quantization

If quantization levels are low:

- image shows **false contours**
- loss of brightness detail

Example: posterized images.

Difference Between Sampling and Quantization

Feature	Sampling	Quantization
Meaning	Converts spatial coordinates to pixels	Converts intensity values to discrete levels

Feature	Sampling	Quantization
Works on	Image position (x, y)	Pixel brightness
Determines	Spatial resolution	Intensity resolution
Example	1024 × 1024 pixels	256 intensity levels

Example of Complete Digitization

Suppose an image has:

- **Resolution:** 512 × 512 pixels
- **Bit depth:** 8 bits

Then:

Total pixels:

$$512 \times 512 = 262,144$$

Intensity levels:

$$2^8 = 256$$

So each pixel stores a value between **0 and 255**.

Steps in Image Processing

Digital Image Processing involves a **series of steps used to analyze, improve, and extract information from images.**

The basic steps are:

1. Image Acquisition
2. Image Enhancement
3. Image Restoration
4. Color Image Processing
5. Wavelets and Multiresolution Processing
6. Image Compression
7. Morphological Processing
8. Image Segmentation
9. Representation & Description
10. Object Recognition

These steps form the **image processing pipeline.**

1. Image Acquisition

This is the **first step** in image processing.

It involves:

- **Capturing an image**
- **Converting it into digital form**

Devices used:

- Digital camera
- Scanner
- Medical imaging devices

Example:

A camera captures a photo and converts it into **pixels stored in a computer.**

Key processes:

- Sampling
 - Quantization
-

2. Image Enhancement

Image enhancement improves the **visual appearance** of an image.

Purpose:

- Make image clearer for **human viewing**

Techniques:

- Contrast adjustment
- Histogram equalization
- Noise reduction
- Brightness correction

Example:

Improving a **dark photograph** so details become visible.

3. Image Restoration

Image restoration focuses on **removing image degradation**.

It uses **mathematical models** to restore the original image.

Common problems:

- Blur
- Noise
- Distortion

Techniques:

- Inverse filtering
- Wiener filtering

Example:

Restoring a **blurred satellite image**.

4. Color Image Processing

This step deals with **processing color images**.

It involves working with color models like:

- RGB
- HSV
- CMY

Applications:

- Image editing
 - Digital photography
 - Video processing
-

5. Wavelets and Multiresolution Processing

This technique analyzes images at **different scales or resolutions**.

It helps to:

- detect details
- compress images

Example:

Wavelet transform used in **JPEG2000 compression**.

6. Image Compression

Image compression reduces the **size of an image file**.

Purpose:

- Save storage space
- Faster transmission

Types:

1. Lossless compression
2. Lossy compression

Examples:

- JPEG
 - PNG
-

7. Morphological Processing

Morphological operations analyze **image shapes and structures**.

Common operations:

- Dilation
- Erosion
- Opening
- Closing

Used in:

- Object detection
 - Shape analysis
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8. Image Segmentation

Segmentation divides an image into **different meaningful regions**.

Goal:

- Identify objects inside an image

Methods:

- Edge detection
- Thresholding
- Clustering

Example:

Separating **tumor region from medical scan**.

9. Representation and Description

After segmentation, objects are represented and described.

Representation:

- Boundary representation
- Region representation

Description (Feature extraction):

- Shape
- Texture
- Size

Example:

Extracting features of a **fingerprint**.

10. Object Recognition

This is the **final step**.

The system identifies objects using extracted features.

Example applications:

- Face recognition
 - Number plate recognition
 - Medical diagnosis
 - AI vision systems
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Summary Flow

Image Acquisition



Image Enhancement



Image Restoration



Color Image Processing



Wavelets & Multiresolution



Image Compression



Morphological Processing



Image Segmentation



Representation & Description



Object Recognition

Image Acquisition

Definition

Image Acquisition is the **first step in digital image processing** where a real-world image is captured and converted into a **digital form** so that a computer can process it.

In simple words:

Image acquisition means **capturing an image using a sensor and converting it into a digital image.**

Basic Components of Image Acquisition System

An image acquisition system mainly consists of **three parts**:

1. **Imaging Sensor**
 2. **Digitizer**
 3. **Processing System**
-

1. Imaging Sensor

The **sensor captures the image** from the real world.

Sensors convert **light energy into electrical signals.**

Common sensors:

- CCD (Charge Coupled Device)
- CMOS sensors

Devices using sensors:

- Digital cameras
- Mobile cameras
- Scanners
- Medical imaging machines

Example:

When you take a photo with your phone, the **camera sensor captures light from the scene.**

2. Digitizer

The electrical signal produced by the sensor is **continuous (analog)**.

The **digitizer converts this analog signal into digital form**.

This process involves:

- **Sampling**
- **Quantization**

After digitization, the image becomes a **matrix of pixels**.

Example:

120 135 140

110 125 130

100 115 120

Each number represents the **intensity value of a pixel**.

3. Processing System

After digitization, the image is sent to the **computer or image processing system**.

The computer performs operations such as:

- Enhancement
 - Filtering
 - Segmentation
 - Recognition
-

Image Acquisition Process

The image acquisition process occurs in the following steps:

1. **Scene illumination**
Light falls on the object.
2. **Reflection from object**
The object reflects the light.

3. **Sensor captures the light**

Camera sensor converts light into electrical signal.

4. **Digitization**

Analog signal is converted into digital form.

5. **Digital image formation**

The image is stored as **pixels in a computer**.

Example

Suppose you take a picture of a **tree using a camera**:

1. Sunlight falls on the tree.
 2. The tree reflects the light.
 3. The camera sensor captures the light.
 4. Sensor converts it into electrical signal.
 5. The digitizer converts the signal into pixels.
 6. The computer stores the image.
-

Types of Image Acquisition Devices

Common devices include:

1. **Digital Cameras**
2. **Scanners**
3. **Medical Imaging Devices** (MRI, CT Scan)
4. **Satellite Cameras**
5. **Microscopes**

Applications:

- Medical imaging
 - Remote sensing
 - Surveillance systems
 - Machine vision
-

Characteristics of Image Acquisition

Important factors affecting image acquisition:

- **Resolution**
- **Lighting conditions**
- **Sensor quality**
- **Noise level**

Better sensors produce **higher quality images**.

Advantages of Image Acquisition

- Allows images to be processed digitally
 - Enables image analysis
 - Used in many fields like **medicine, security, and AI**
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Color Image Representation

Definition

Color Image Representation is the method of representing a digital image using **color information instead of only brightness (intensity)**.

A color image is formed by **combining multiple color components** to produce different colors.

In digital image processing, color images are usually represented using **color models**.

Components of Color Image

A color image is made up of **three basic color components**:

- **Red (R)**
- **Green (G)**
- **Blue (B)**

These three components combine to form **different colors**.

Example pixel:

R G B

255 0 0

This pixel represents **pure red color**.

RGB Color Model

The most common method for representing color images is the **RGB model**.

In this model:

- Each pixel contains **three values**:
 - Red intensity
 - Green intensity
 - Blue intensity

Each value usually ranges from:

0 to 255

Example:

R	G	B	Color
---	---	---	-------

255	0	0	Red
-----	---	---	-----

0	255	0	Green
---	-----	---	-------

0	0	255	Blue
---	---	-----	------

255	255	255	White
-----	-----	-----	-------

0	0	0	Black
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Matrix Representation of Color Image

A color image can be represented by **three matrices**:

1. Red Matrix
2. Green Matrix
3. Blue Matrix

Example:

Red Channel

255	200
-----	-----

150	100
-----	-----

Green Channel

50	60
----	----

70	80
----	----

Blue Channel

20	40
----	----

60	90
----	----

These matrices combine to create the **final color image**.

Bit Depth of Color Images

Color images usually use **24-bit representation**.

Explanation:

- 8 bits for **Red**
- 8 bits for **Green**
- 8 bits for **Blue**

Total:

$$8 + 8 + 8 = 24 \text{ bits}$$

Total possible colors:

$$256^3 = 16,777,216$$

So a 24-bit image can display **over 16 million colors**.

Other Color Models

Besides RGB, other color models are also used.

1. CMY Model

Used in **printing systems**.

Colors:

- Cyan
 - Magenta
 - Yellow
-

2. HSV Model

Used in **image editing and computer graphics**.

Components:

- **H (Hue)** → color type
 - **S (Saturation)** → color purity
 - **V (Value)** → brightness
-

3. YIQ / YCbCr Model

Used in **television broadcasting and video compression**.

Applications of Color Image Representation

Color image representation is used in:

- Digital photography
 - Video processing
 - Computer vision
 - Medical imaging
 - Satellite imaging
 - Graphic design
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Advantages

- Provides **more information than grayscale images**
- Helps in **object detection and recognition**
- Improves **visual quality**
-

Unit = 2

Intensity Transform Functions

Definition

Intensity Transform Functions are mathematical operations used to **modify the intensity (brightness) values of pixels in an image**.

The goal is to **enhance the image quality or highlight important details**.

In intensity transformation, each pixel value **r** is transformed into a new value **s** using a transformation function.

Mathematical Representation

Intensity transformation is represented as:

$$s = T(r)$$

Where:

- **r** = input pixel intensity
- **s** = output pixel intensity
- **T** = transformation function

The transformation is applied **independently to every pixel**.

Types of Intensity Transformations

1. Image Negative

Image negative is obtained by **reversing the intensity values**.

Formula:

$$s = L - 1 - r$$

Where:

- **L** = maximum intensity level
- **r** = original pixel value

Example for **8-bit image**:

$$L = 256$$

So:

$$s = 255 - r$$

Example:

Original Pixel Negative Pixel

0	255
50	205
200	55

Uses:

- Medical imaging
- Enhancing white or gray details on dark backgrounds

2. Log Transformation

Log transformation is used to **expand dark pixels and compress bright pixels**.

Formula:

$$s = c \log (1 + r)$$

Where:

- **c** = constant
- **r** = input pixel value

Features:

- Enhances **low intensity regions**
- Compresses **high intensity regions**

Applications:

- Fourier spectrum visualization
- Image enhancement

3. Power-Law Transformation (Gamma Correction)

Power-law transformation controls **brightness using gamma value**.

Formula:

$$s = cr^\gamma$$

Where:

- **c** = constant
- **γ (gamma)** = power value

Cases:

Gamma Value Effect

$\gamma < 1$ Image becomes brighter

$\gamma > 1$ Image becomes darker

Applications:

- Display monitors
- Television image correction

4. Piecewise Linear Transformation

This transformation modifies pixel intensity using **multiple linear segments**.

Types include:

- Contrast stretching
- Gray level slicing
- Bit-plane slicing

These techniques help **highlight specific intensity ranges**.

(a) Contrast Stretching

Used to **increase the contrast of an image**.

It expands the intensity range.

Example:

Original range:

20 – 150

After stretching:

0 – 255

Result → clearer image.

(b) Gray Level Slicing

Highlights a **specific range of gray levels**.

Example:

Highlight pixel values between **100–150**.

Used in:

- Medical imaging
 - Satellite images
-

(c) Bit-Plane Slicing

Each pixel is represented by **bits**.

Example:

8-bit pixel

10101100

Each bit plane can be separated to analyze image details.

Used in:

- Image compression
 - Feature extraction
-

Advantages of Intensity Transformations

- Improves image visibility
- Enhances contrast

- Highlights important details
 - Removes unnecessary intensity variations
-

Applications

- Medical imaging
 - Satellite image processing
 - Computer vision
 - Digital photography
-

Histogram Processing

Definition

Histogram Processing is a technique used in digital image processing to **analyze and improve the contrast of an image** by studying the distribution of pixel intensity values.

A **histogram** is a **graph that shows how many pixels in an image have each intensity value**.

Image Histogram

An **image histogram** represents the **frequency of occurrence of gray levels** in an image.

For a grayscale image:

- **X-axis:** Intensity levels (0–255)
- **Y-axis:** Number of pixels with that intensity

Example:

Intensity Value Number of Pixels

0	50
50	120
100	200
150	300
200	180

This table represents the **histogram distribution**.

Mathematical Representation

Let:

- r_k = intensity level
- n_k = number of pixels with intensity r_k
- n = total number of pixels in the image

Histogram probability:

$$p(r_k) = \frac{n_k}{n}$$

Where:

- $p(r_k)$ = probability of intensity level r_k
-

Types of Histogram Shapes

1. Dark Image

Histogram concentrated on **left side**.

Meaning:

- Most pixel values are **low intensity**

Result → image appears **too dark**.

2. Bright Image

Histogram concentrated on **right side**.

Meaning:

- Most pixel values are **high intensity**

Result → image appears **too bright**.

3. Low Contrast Image

Histogram concentrated in **narrow middle range**.

Meaning:

- Pixel values are very similar

Result → **dull image**.

4. High Contrast Image

Histogram spread across **entire intensity range (0–255)**.

Result → **clear and detailed image**.

Histogram Processing Techniques

1. Histogram Equalization

Histogram equalization is used to **improve image contrast** by spreading pixel intensities across the full range.

Purpose:

- Make histogram **more uniform**

Effect:

- Dark regions become visible
- Overall image contrast improves

Steps:

1. Compute histogram
2. Normalize histogram
3. Compute cumulative distribution
4. Map old pixel values to new values

Applications:

- Medical imaging
 - Satellite images
 - Face recognition
-

2. Histogram Specification (Matching)

Histogram specification transforms the histogram of an image to **match a desired histogram shape**.

Used when we want **specific contrast characteristics**.

Example:

- Matching brightness levels of two images.
-

Advantages of Histogram Processing

- Improves **image contrast**
- Enhances **visual quality**

- Reveals hidden details
 - Simple and fast method
-

Applications

Histogram processing is used in:

- Medical imaging
 - Satellite image analysis
 - Computer vision
 - Security systems
 - Image enhancement
-

Example

Original image:

Low contrast → most pixel values between **80–120**.

After histogram equalization:

Pixel values spread between **0–255**.

Result → **sharper and clearer image**.

Spatial Filtering

Definition

Spatial Filtering is a technique in digital image processing where the **pixel value is modified based on the values of its neighboring pixels**.

It is used to **enhance an image, remove noise, or detect edges**.

In spatial filtering, operations are performed **directly on the image pixels in the spatial domain**.

Basic Idea

Spatial filtering uses a **small matrix called a mask, kernel, or filter**.

This mask moves across the image and performs calculations with neighboring pixels.

Example mask (3×3):

1 1 1

1 1 1

1 1 1

This mask interacts with the **pixels under it**.

Spatial Filtering Operation

Steps:

1. Place the **mask on the image**.
2. Multiply mask values with corresponding pixel values.
3. Add all the results.
4. Replace the center pixel with the computed value.

Example:

Image section:

100 110 120

90 100 110

100 110 120

80 90 100

Mask:

1 1 1

1 1 1

1 1 1

Result = **average of neighboring pixels.**

Types of Spatial Filtering

There are **two main types**:

1. **Smoothing Filters**
 2. **Sharpening Filters**
-

1. Smoothing Filters

Smoothing filters are used to **reduce noise and blur the image slightly.**

They remove **small variations in intensity.**

Types:

(a) Average Filter

Each pixel value is replaced by the **average of its neighboring pixels.**

Example mask:

1/9 1/9 1/9

1/9 1/9 1/9

1/9 1/9 1/9

Effect:

- Noise reduction
 - Image becomes smoother.
-

(b) Median Filter

The median filter replaces the pixel value with the **median value of neighboring pixels**.

Example:

Pixel values:

10 20 30

40 50 60

70 80 200

Sorted values:

10 20 30 40 50 60 70 80 200

Median = **50**

Used for:

- **Salt-and-pepper noise removal**
-

2. Sharpening Filters

Sharpening filters are used to **highlight edges and fine details** in an image.

They increase **contrast at boundaries**.

(a) Laplacian Filter

Used for **edge enhancement**.

Example mask:

0 -1 0

-1 4 -1

0 -1 0

This filter detects **rapid intensity changes**.

(b) High-Pass Filter

High-pass filters enhance **high-frequency components** like edges and textures.

Example mask:

-1 -1 -1

-1 8 -1

-1 -1 -1

Result → **sharper image.**

Advantages of Spatial Filtering

- Improves image quality
- Removes noise
- Enhances edges
- Simple to implement

Applications

Spatial filtering is used in:

- Medical image analysis
- Face detection
- Satellite image processing
- Computer vision
- Pattern recognition

Difference Between Smoothing and Sharpening Filters

Feature	Smoothing Filter	Sharpening Filter
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Purpose	Reduce noise	Highlight edges
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Effect	Blur image slightly	Increase image detail
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Example	Average filter	Laplacian filter
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Fourier Transform

Definition

The **Fourier Transform (FT)** is a mathematical technique used to **convert an image from the spatial domain into the frequency domain**.

- **Spatial domain:** image represented by pixels $f(x, y)$
- **Frequency domain:** image represented by frequencies $F(u, v)$

It helps analyze **how rapidly pixel values change in an image**.

Mathematical Representation

The **2D Fourier Transform** of an image is:

$$F(u, v) = \sum_{x=0}^{M-1} \sum_{y=0}^{N-1} f(x, y) e^{-j2\pi \left(\frac{ux}{M} + \frac{vy}{N} \right)}$$

Where:

- $f(x, y)$ = original image
 - $F(u, v)$ = frequency representation
 - M, N = image size
 - u, v = frequency coordinates
 - $e^{-j2\pi(\cdot)}$ = complex exponential function
-

Idea Behind Fourier Transform

An image can be considered as a **combination of many sinusoidal waves (frequencies)**.

Fourier Transform separates these frequencies.

Types of Frequencies

Low Frequency

- Represents **smooth areas**
- Slow intensity changes
- Contains general shape of image

High Frequency

- Represents **edges and fine details**
- Rapid intensity changes

Example

Original Image

→ Converted to **frequency spectrum**

- Center of spectrum → **low frequencies**
- Edges of spectrum → **high frequencies**

Inverse Fourier Transform

After processing the image in the frequency domain, we convert it back to spatial form using the **Inverse Fourier Transform**.

$$f(x,y) = \frac{1}{MN} \sum_{u=0}^{M-1} \sum_{v=0}^{N-1} F(u,v) e^{j2\pi \left(\frac{ux}{M} + \frac{vy}{N} \right)}$$

This reconstructs the **original image**.

Properties of Fourier Transform

1. Linearity Property

If two images are combined, their Fourier transforms also combine linearly.

If:

$$f(x,y) = af_1(x,y) + bf_2(x,y)$$

Then:

$$F(u,v) = aF_1(u,v) + bF_2(u,v)$$

Meaning → transform of sum = sum of transforms.

2. Translation (Shift) Property

Shifting an image in the spatial domain causes a **phase shift in the frequency domain**.

If the image shifts by x_0, y_0 :

$$f(x - x_0, y - y_0)$$

The magnitude remains same but **phase changes**.

3. Scaling Property

Scaling the image size changes the frequency components.

- **Large image structures → low frequencies**
 - **Small details → high frequencies**
-

4. Rotation Property

If an image is rotated in the spatial domain, its Fourier transform also rotates by the **same angle** in the frequency domain.

5. Convolution Property

Convolution in the spatial domain becomes **multiplication in the frequency domain**.

This makes filtering easier.

Example:

$$f(x, y) * h(x, y) \leftrightarrow F(u, v)H(u, v)$$

Where:

- $h(x, y)$ = filter
 - $H(u, v)$ = filter in frequency domain
-

6. Symmetry Property

For real images, the Fourier transform shows **symmetry**.

This reduces computational complexity.

Applications of Fourier Transform

Fourier Transform is widely used in:

- Image filtering
 - Noise removal
 - Image compression
 - Medical imaging
 - Pattern recognition
 - Satellite image analysis
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Advantages

- Separates image frequencies
 - Helps in frequency-based filtering
 - Useful for image enhancement
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Frequency Domain Filters

Definition

Frequency Domain Filtering is a technique in digital image processing where an image is **processed in the frequency domain to enhance or suppress certain frequency components**.

Instead of working directly on pixels, the image is first converted to the **frequency domain using the Fourier Transform**, then filtering is applied.

Basic Steps of Frequency Domain Filtering

1. Convert the image $f(x, y)$ into frequency domain using **Fourier Transform**.
2. Multiply the frequency image with a **filter function** $H(u, v)$.
3. Apply **Inverse Fourier Transform** to obtain the processed image.

Mathematically:

$$G(u, v) = H(u, v) \times F(u, v)$$

Where:

- $F(u, v)$ = Fourier Transform of the image
- $H(u, v)$ = Filter function
- $G(u, v)$ = Filtered image in frequency domain

After applying inverse transform, we get the final image $g(x, y)$.

Types of Frequency Domain Filters

Frequency filters are mainly divided into:

1. **Low Pass Filters**
 2. **High Pass Filters**
 3. **Band Pass Filters**
 4. **Band Reject Filters**
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1. Low Pass Filter (LPF)

A **Low Pass Filter** allows **low frequencies to pass** and blocks **high frequencies**.

Effect:

- Smooths the image
- Removes noise
- Reduces sharp edges

Example:

Blurred images are produced using LPF.

Types of LPF:

- Ideal Low Pass Filter
- Butterworth Low Pass Filter
- Gaussian Low Pass Filter

Applications:

- Noise removal
 - Image smoothing
-

2. High Pass Filter (HPF)

A **High Pass Filter** allows **high frequencies to pass** and removes **low frequencies**.

Effect:

- Enhances edges
- Highlights fine details

Example:

Edge detection in images.

Types:

- Ideal High Pass Filter
- Butterworth High Pass Filter
- Gaussian High Pass Filter

Applications:

- Image sharpening

- Edge detection
-

3. Band Pass Filter (BPF)

A **Band Pass Filter** allows only a **specific range of frequencies** to pass.

Effect:

- Removes both very low and very high frequencies.

Applications:

- Pattern recognition
 - Texture analysis
-

4. Band Reject Filter (BRF)

A **Band Reject Filter** blocks a **specific frequency range** and allows others to pass.

Also called:

Notch Filter

Applications:

- Removing periodic noise
 - Signal interference removal
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Frequency Spectrum Concept

In a frequency spectrum:

- **Center** → Low frequencies
- **Edges** → High frequencies

Filters work by **modifying these frequency regions**.

Advantages of Frequency Domain Filtering

- Effective for **noise removal**
- Easier to implement large filters
- Useful for **image enhancement**

Applications

Frequency domain filters are used in:

- Medical image processing
 - Satellite imaging
 - Pattern recognition
 - Image restoration
 - Computer vision
-

Colour Models

Definition

A **Colour Model** is a mathematical model used to **represent colors using numerical values**.

It defines how colors are **represented and displayed in digital images**.

Different applications use **different color models**.

Example:

- Computer displays → RGB
 - Printing → CMY/CMYK
 - Image processing → HSV, HSI
-

Basic Idea of Colour Models

Most color models represent colors using **three components**.

Example:

- Red
- Green
- Blue

These components combine to produce **millions of colors**.

1. RGB Colour Model

The **RGB model** is the most commonly used color model in **digital displays**.

Components:

- **R** → Red
- **G** → Green
- **B** → Blue

Each component typically ranges from:

0 to 255

Example:

R	G	B	Color
---	---	---	-------

255	0	0	Red
-----	---	---	-----

0	255	0	Green
---	-----	---	-------

0	0	255	Blue
---	---	-----	------

255	255	255	White
-----	-----	-----	-------

0	0	0	Black
---	---	---	-------

Total possible colors:

$$256^3 = 16,777,216$$

Applications:

- Computer monitors
 - Television screens
 - Mobile displays
 - Digital cameras
-

2. CMY Colour Model

The **CMY model** is mainly used in **printing systems**.

Components:

- **C** → Cyan
- **M** → Magenta
- **Y** → Yellow

This model is the **complement of RGB**.

Relationship with RGB:

$$C = 1 - R$$

$$M = 1 - G$$

$$Y = 1 - B$$

Example:

C M Y Color

0 1 1 Red

1 0 1 Green

1 1 0 Blue

Printers often use **CMYK**, where:

K = Black

3. HSV Colour Model

HSV stands for:

- **H → Hue**
- **S → Saturation**
- **V → Value (Brightness)**

This model represents colors in a way **closer to human perception**.

Hue

Represents the **type of color**.

Example:

- 0° → Red
- 120° → Green
- 240° → Blue

Saturation

Represents **color purity**.

- 0 → gray
- 1 → pure color

Value

Represents **brightness**.

Range:

0 → black

1 → brightest color

Applications:

- Image editing
 - Computer graphics
 - Object detection
-

4. HSI Colour Model

HSI stands for:

- **Hue**
- **Saturation**
- **Intensity**

It is similar to HSV but separates **brightness (intensity)** from color.

Used in:

- Image analysis
 - Computer vision
-

Comparison of Colour Models

Model Components		Used In
RGB	Red, Green, Blue	Displays & cameras
CMY	Cyan, Magenta, Yellow	Printing
HSV	Hue, Saturation, Value	Graphics & editing
HSI	Hue, Saturation, Intensity	Image analysis

Applications of Colour Models

Colour models are used in:

- Digital photography
- Video processing
- Computer graphics

- Medical imaging
 - Object detection
 - Satellite imaging
-

Advantages

- Represent millions of colors
 - Useful for image enhancement
 - Helps in color-based segmentation
-

Pseudo Colouring

Definition

Pseudo Colouring (False Colouring) is a technique used to **assign colors to grayscale images** to make them easier to analyze and interpret.

In a grayscale image, pixel values represent **only intensity levels**.

Pseudo colouring converts these intensity values into **different colors**.

This helps **highlight important features** in the image.

Basic Idea

A grayscale image contains intensity values ranging from:

0 → 255

Pseudo colouring maps these intensity values to **specific colors**.

Example:

Intensity Value Assigned Color

0–50	Blue
51–100	Green
101–150	Yellow
151–200	Orange
201–255	Red

So instead of **gray shades**, we see **different colors**.

Need for Pseudo Colouring

Human eyes can **distinguish colors better than gray levels**.

Pseudo colouring helps to:

- Highlight details
 - Improve visualization
 - Make patterns easier to understand
-

Methods of Pseudo Colouring

There are mainly **two methods**:

1. **Intensity Slicing (Density Slicing)**
2. **Intensity to Color Transformation**

1. Intensity Slicing (Density Slicing)

In this method, the grayscale intensity range is **divided into several intervals**.

Each interval is assigned a **specific color**.

Example:

Intensity Range Color

0–60	Blue
61–120	Green
121–180	Yellow
181–255	Red

Applications:

- Medical imaging
- Satellite images

2. Intensity to Colour Transformation

In this method, intensity values are mapped to **RGB color components** using mathematical functions.

Each grayscale value generates a **unique color**.

Example:

If intensity increases:

Blue → Green → Yellow → Red

This produces a **smooth color transition**.

Example

Grayscale Image:

50 80 120
150 180 200
210 230 255

After pseudo colouring:

Blue Green Yellow
Yellow Orange Red
Red Red Red

The image becomes **more informative and easier to analyze**.

Applications of Pseudo Colouring

Pseudo colouring is widely used in:

- **Medical imaging** (X-ray, MRI)
- **Satellite image analysis**
- **Thermal imaging**
- **Weather analysis**
- **Remote sensing**

Example:

Temperature maps often use **red for hot regions and blue for cold regions**.

Advantages

- Improves visualization
 - Highlights image features
 - Helps in data interpretation
 - Easy to implement
-

Disadvantages

- Colors are **artificially assigned**
- May not represent real-world colors

Colour Transforms

Definition

Colour Transform is the process of **converting an image from one color model to another color model**.

It helps in **image enhancement, compression, and analysis**.

Example:

- RGB → HSV
 - RGB → YCbCr
 - RGB → HSI
-

Need for Colour Transforms

Different applications require **different color models**.

Reasons for color transformation:

- Improve image processing efficiency
 - Separate brightness from color information
 - Simplify image analysis
 - Improve compression techniques
-

Common Colour Transformations

1. RGB to CMY Transformation

This transformation is commonly used in **printing systems**.

Relationship between RGB and CMY:

$$\begin{aligned}C &= 1 - R \\M &= 1 - G \\Y &= 1 - B\end{aligned}$$

Where:

- **R, G, B** = Red, Green, Blue
- **C, M, Y** = Cyan, Magenta, Yellow

Example:

If

$R = 0.2$

$G = 0.5$

$B = 0.7$

Then

$C = 0.8$

$M = 0.5$

$Y = 0.3$

2. RGB to HSV Transformation

HSV stands for:

- **Hue**
- **Saturation**
- **Value**

Meaning:

Hue → type of color

Saturation → purity of color

Value → brightness

This transformation helps in **object detection and color segmentation**.

Applications:

- Image editing
- Computer graphics
- Object tracking

3. RGB to YCbCr Transformation

This transform separates **luminance and chrominance components**.

Components:

- **Y** → luminance (brightness)
- **Cb** → blue chrominance

- **Cr** → red chrominance

Used in:

- Digital video processing
- Image compression (JPEG, MPEG)

Advantage:

Brightness information is separated from color information.

4. RGB to HSI Transformation

HSI stands for:

- **Hue**
- **Saturation**
- **Intensity**

This model closely represents **human perception of colors**.

Applications:

- Image enhancement
 - Image segmentation
 - Computer vision
-

Example of Colour Transformation

Suppose an image is stored in **RGB format**.

For image compression:

RGB → **YCbCr**

For image editing:

RGB → **HSV**

For printing:

RGB → **CMY**

Advantages of Colour Transforms

- Makes image processing easier
 - Improves color analysis
 - Useful for compression
 - Helps in object detection
-

Applications

Colour transforms are used in:

- Computer vision
- Image compression
- Video processing
- Face detection
- Satellite image analysis

Basics of Wavelet Transform

Definition

The **Wavelet Transform** is a mathematical technique used to **analyze an image at different scales or resolutions**.

It breaks an image into **small wave-like components called wavelets**.

Unlike the Fourier transform (which analyzes only frequency), the wavelet transform analyzes **both frequency and location** of features in an image.

Basic Idea

An image contains:

- **Low-frequency components** → smooth areas
- **High-frequency components** → edges and fine details

Wavelet transform separates these components so that the image can be **analyzed more efficiently**.

Wavelet

A **wavelet** is a **small wave with limited duration**.

Characteristics:

- Short in time or space
- Oscillatory (wave-like shape)
- Localized

Wavelets are used as **basis functions** to represent images.

Wavelet Transform Process

The wavelet transform works by **decomposing the image into sub-images**.

Main steps:

1. Apply **low-pass filter** → produces approximation (low frequency)
2. Apply **high-pass filter** → produces detail components
3. Downsample the results

This creates different **sub-bands of the image**.

Sub-band Decomposition

After wavelet transform, the image is divided into **four parts**:

Component Meaning

LL	Approximation (low frequency)
LH	Horizontal details
HL	Vertical details
HH	Diagonal details

Explanation:

- **LL** → main structure of image
 - **LH** → horizontal edges
 - **HL** → vertical edges
 - **HH** → diagonal edges
-

Multi-Resolution Analysis

Wavelet transform allows **multi-resolution analysis**.

This means the image can be analyzed at **different levels of detail**.

Example:

Level 1 → basic features

Level 2 → finer details

Level 3 → very fine details

This is useful in **image compression and feature extraction**.

Types of Wavelet Transforms

1. Continuous Wavelet Transform (CWT)

- Uses continuous scaling and shifting
- Mostly used in **signal analysis**

2. Discrete Wavelet Transform (DWT)

- Uses discrete values
- Most commonly used in **image processing**

Applications of Wavelet Transform

Wavelet transform is widely used in:

- Image compression (JPEG2000)
- Image denoising
- Feature extraction
- Medical image processing
- Satellite image analysis

Advantages

- Provides **multi-resolution analysis**
- Detects **edges and textures effectively**
- Works well for **image compression**

Difference Between Fourier Transform and Wavelet Transform

Feature	Fourier Transform	Wavelet Transform
Analysis	Only frequency	Frequency + location
Localization	Poor	Good
Best for	Stationary signals	Images and non-stationary signals

Unit = 3

Image Degradation and Restoration Process

Definition

Image Degradation refers to the **loss of image quality** caused by factors such as noise, blur, or distortion during image acquisition or transmission.

Image Restoration is the process of **recovering the original image by removing or reducing degradation** using mathematical techniques.

Image Degradation Model

The degradation process is usually represented using a mathematical model.

The degraded image can be expressed as:

$$g(x, y) = h(x, y) * f(x, y) + n(x, y)$$

Where:

- **f(x,y)** → original image
- **h(x,y)** → degradation function (blur)
- **n(x,y)** → noise
- **g(x,y)** → degraded image
- ***** → convolution operation

Meaning:

Original image → affected by blur and noise → degraded image.

Causes of Image Degradation

Images can degrade due to several factors:

1. Noise

Random variations in pixel intensity.

Examples:

- Gaussian noise
- Salt and pepper noise

2. Blur

Occurs due to:

- Camera motion
 - Out-of-focus lens
 - Atmospheric turbulence
-

3. Transmission Errors

Data loss during **image transmission over networks**.

4. Sensor Limitations

Poor quality sensors can introduce **distortion and noise**.

Image Restoration

Image restoration aims to **recover the original image** using knowledge about the degradation process.

It uses **mathematical models and filtering techniques**.

Goal:

Remove noise and blur to obtain the **best possible estimate of the original image**.

Image Restoration Process

The restoration process involves the following steps:

1. Identify degradation model

Determine how the image was degraded.

2. Estimate degradation function

Find the function $h(x, y)$ causing blur or distortion.

3. Apply restoration filter

Use techniques like:

- Inverse filtering
- Wiener filtering

4. Reconstruct the image

The processed image becomes closer to the **original image**.

Difference Between Image Enhancement and Restoration

Feature	Image Enhancement	Image Restoration
Goal	Improve visual appearance	Recover original image
Method	Heuristic techniques	Mathematical models
Example	Increase contrast	Remove blur

Applications

Image degradation and restoration are used in:

- Medical imaging
 - Satellite imaging
 - Astronomy images
 - Surveillance systems
 - Old photo restoration
-

Advantages of Image Restoration

- Improves image quality
 - Recovers lost information
 - Useful in scientific imaging
-

Noise Models

Definition

Noise in digital images refers to **random variations in pixel intensity values** that degrade the quality of the image.

A **Noise Model** mathematically describes how noise affects an image.

Noise usually occurs during:

- Image acquisition
 - Image transmission
 - Sensor errors
 - Environmental conditions
-

Mathematical Representation

A noisy image can be represented as:

$$g(x, y) = f(x, y) + n(x, y)$$

Where:

- **f(x,y)** → original image
- **n(x,y)** → noise
- **g(x,y)** → noisy image

Meaning:

Original image + Noise = Noisy image

Types of Noise Models

1. Gaussian Noise

Gaussian noise is the **most common noise model** in digital images.

It follows a **normal distribution**.

Probability density function:

$$p(z) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(z-\mu)^2}{2\sigma^2}}$$

Where:

- μ (**mu**) = mean
- σ^2 (**sigma**²) = variance

Characteristics:

- Appears as **random brightness variations**
- Found in **electronic circuits and sensors**

Applications:

- Medical imaging
 - Camera sensors
-

2. Salt and Pepper Noise

Salt and Pepper noise appears as **random black and white pixels** in an image.

Pixel values suddenly become:

- **0 → black (pepper)**
- **255 → white (salt)**

Example:

120 125 130
0 128 255
115 120 125

Cause:

- Transmission errors
- Faulty sensors

Removal method:

- **Median filtering**
-

3. Uniform Noise

Uniform noise has **equal probability for all intensity values** within a range.

Probability density function:

$$p(z) = \begin{cases} \frac{1}{b-a}, & a \leq z \leq b \\ 0, & \text{otherwise} \end{cases}$$

Where:

- **a, b** → intensity range

Characteristics:

- Noise values are evenly distributed.
-

4. Rayleigh Noise

Rayleigh noise occurs in **radar imaging and remote sensing**.

Probability density function:

$$p(z) = \begin{cases} \frac{z-a}{b} e^{-\frac{(z-a)^2}{2b}}, & z \geq a \\ 0, & z < a \end{cases}$$

Characteristics:

- Used to model **random scattering effects**.
-

5. Exponential Noise

Exponential noise follows an **exponential distribution**.

Probability density function:

$$p(z) = \begin{cases} ae^{-az}, & z \geq 0 \\ 0, & z < 0 \end{cases}$$

Where:

- **a** is a constant.

Applications:

- Signal processing systems.
-

6. Gamma (Erlang) Noise

Gamma noise is used to describe **multiplicative noise in imaging systems**.

Probability density function:

$$p(z) = \begin{cases} \frac{a^b z^{b-1} e^{-az}}{(b-1)!}, & z \geq 0 \\ 0, & z < 0 \end{cases}$$

Where:

- **a, b** are parameters.
-

Effects of Noise on Images

Noise can cause:

- Loss of image details
 - Blurred images
 - False edges
 - Poor image quality
-

Methods to Remove Noise

Noise can be reduced using filters such as:

- Mean filter
 - Median filter
 - Gaussian filter
 - Wiener filter
-

Applications

Noise models are important in:

- Image restoration
- Medical imaging
- Satellite images
- Computer vision

Noise Filters

Definition

Noise Filters are techniques used in digital image processing to **remove or reduce noise from an image while preserving important details.**

Noise filters are commonly used in the **image restoration process.**

Goal:

- Reduce noise
 - Improve image quality
 - Preserve edges and details
-

Types of Noise Filters

Noise filters are mainly divided into **two categories:**

1. **Linear Filters**
 2. **Non-Linear Filters**
-

1. Linear Noise Filters

Linear filters remove noise by **performing mathematical operations on neighboring pixels.**

The output pixel is obtained using a **linear combination of neighboring pixels.**

Common linear filters:

(a) Mean Filter (Average Filter)

The mean filter replaces each pixel value with the **average of its neighboring pixels.**

Example mask:

1/9 1/9 1/9

1/9 1/9 1/9

1/9 1/9 1/9

Effect:

- Reduces **Gaussian noise**

- Smooths the image

Disadvantage:

- Can **blur edges and details**
-

(b) Gaussian Filter

The Gaussian filter uses a **Gaussian distribution** to smooth the image.

Characteristics:

- Gives **more weight to central pixels**
- Produces **smooth noise reduction**

Applications:

- Image smoothing
 - Preprocessing in computer vision
-

2. Non-Linear Noise Filters

Non-linear filters process pixel values using **non-linear operations**.

They are **more effective for removing certain types of noise**.

(a) Median Filter

The median filter replaces a pixel with the **median value of neighboring pixels**.

Example:

Pixel values:

10 20 30

40 50 60

70 80 200

Sorted values:

10 20 30 40 50 60 70 80 200

Median value = **50**

Advantages:

- Removes **salt and pepper noise**

- Preserves edges

Applications:

- Medical images
 - Satellite images
-

(b) Max Filter

The max filter replaces the center pixel with the **maximum value in the neighborhood**.

Purpose:

- Removes **pepper noise (black pixels)**
-

(c) Min Filter

The min filter replaces the center pixel with the **minimum value in the neighborhood**.

Purpose:

- Removes **salt noise (white pixels)**
-

Comparison of Noise Filters

Filter	Type	Used For
Mean Filter	Linear	Gaussian noise
Gaussian Filter	Linear	Image smoothing
Median Filter	Non-linear	Salt & pepper noise
Max Filter	Non-linear	Pepper noise
Min Filter	Non-linear	Salt noise

Applications of Noise Filters

Noise filters are used in:

- Medical imaging
- Satellite image processing

- Computer vision systems
 - Surveillance systems
 - Image restoration
-

Advantages

- Improves image quality
 - Removes unwanted noise
 - Enhances visual clarity
-

Disadvantages

- Some filters **blur edges**
 - Important details may be lost if filtering is excessive
-